

TEMPLATE

KEY PROJECT INFORMATION & VPA DESIGN DOCUMENT (PDD)

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VERSION **v. 2.0**

RELATED SUPPORT - [Programme of Activity requirements](#)

This document contains the following Sections

Key Project Information

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KEY PROJECT INFORMATION

Type of VPA	☐ Real case VPA ☒ Regular VPA
Scale of VPA Note that a VPA can be of one scale. Please select applicable scale accordingly.	☐ Microscale ☐ Small scale ☒ Large scale
Title of corresponding real case VPA (if applicable)	NA
GS ID of real case VPA (if applicable)	NA
GS ID of VPA	GS11432
Title of VPA	GS10789 VPA4: Efficient and Clean Cooking for households in Kenya
Time of First Submission Date	01/11/2021
Date of Design Certification	30/12/2022
Version number of the VPA-DD	2.8
Completion date of version	15/11/2023
Coordinating/managing entity	BURN Manufacturing Co.
VPA Implementer (s)	BURN Manufacturing Co.
Project Participants and any communities involved	BURN Manufacturing Co.
Host Country (ies)	Republic of Kenya
GS ID and Title of applicable Design Certified VPA	NA
GS ID and Title of applicable Performance Certified VPA	NA
Activity Requirements applied	☒ Community Services Activities ☐ Renewable Energy Activities

	☐ Land Use and Forestry Activities/Risks & Capacities ☐ N/A
Other Requirements applied	NA
Methodology (ies) applied and version number	Technologies and Practices to Displace Decentralized Thermal Energy Consumption' (TPDDTEC), version 03.1
Product Requirements applied	☒ GHG Emissions Reduction & Sequestration ☐ Renewable Energy Label ☐ N/A
VPA Cycle:	☐ Regular ☒ Retroactive

Table 1 – Estimated Sustainable Development Contributions

Sustainable Development Goals Targeted		SDG Impact (defined in B.6)		Estimated Annual Average	Units or Products
13	Climate (mandatory)	Action	Emission Reductions	171,888	VERs
1	End poverty in all its forms everywhere	Monetary related purchase	savings to the of charcoal	50.89%	Monetary savings in %
3	Ensure healthy lives and promote well-being for all at all ages		Perceived air quality	98.76%	Households in % perceiving improved air quality
7.	Ensure access to affordable, reliable, sustainable, and modern energy for all	Number sold/distributed	of	44,877	ICS in use
8.	Promote sustained, inclusive, and sustainable economic growth, full and	Number created	of jobs	25	Number of jobs created

productive employment and
decent work for all

SECTION A. DESCRIPTION OF PROJECT

A.1. Purpose and general description of project

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This VPA deploys highly efficient improved charcoal cookstoves (ICS), known as 'Jikokoas' reducing woody biomass consumption for urban and peri-urban households in the Republic of Kenya. For a more detailed technical description of the ICS see section A.3. of this document.

The VPA is implemented by BURN Manufacturing Co. (in the following 'BURN'), at the same time Coordinating and Managing Entity (CME) of the PoA, the biggest manufacturer of highly efficient improved cookstoves in Sub-Saharan Africa producing all its stoves in the first and only modern cookstove manufacturing facility in Kenya. BURN collaborates with various local partners on the ground to assist in the different activities of this cookstove project.

The scenario existing prior to the implementation of the project is the use of very inefficient traditional cookstoves, consuming a lot of non-renewable charcoal and firewood. The high biomass consumption has negative impacts on the environment leading to deforestation and land degradation, Greenhouse Gas Emissions (GHG) emissions, loss of soil fertility and soils' reduced ability of water retention. Further, indoor air pollution through health-damaging pollutants while combusting firewood and charcoal result in diseases like e.g. pneumonia, stroke, ischaemic heart diseases, chronic obstructive pulmonary diseases and lung cancer. The project activity will relieve these negative impacts.

The PoA is a voluntary action by the CME.

A.1.1. Eligibility of the VPA under approved PoA

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Table 2 Eligibility for VPA inclusion as per PoA requirements

No.	Eligibility Criterion	Description/ Required condition	Description of the VPA in relation to the criteria,
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			Means of Verification and Supporting evidence for inclusion
1	Geographical Boundary	ICS distributed ¹ under any of the VPAs will be located in any of the countries mentioned under Table 1 of the PoA-DD.	ICS are distributed to urban and peri-urban households all over the territory of Kenya. For more details see section A.2. of this document.
2	Double-counting project activities	of All VPAs will be checked to prevent double counting and are not registered as a separate GS project activity, nor included as part of another registered GS (or other carbon standard) PoA nor that the project activity has been deregistered.	All carbon standard registries (UNFCCC, GS and VERRA) have been checked and it is confirmed that the VPA has not been registered as a separate GS project activity, nor included as part of another registered GS (or other carbon standard) PoA nor that the project activity has been deregistered. The same has been confirmed by a letter signed by the VPA implementer and submitted to GS.
3	Technology	Each VPA will implement improved biomass cook stoves.	The VPA implements highly efficient cookstoves known as 'Jikokoas'. Detailed manufacturer's technology specifications are listed in section A.3 of this document.
4	Conditions to check the start date of the VPA	The start date of a project activity is the date on	The start date is defined as 01/12/2020, i.e. the

¹ Distributed¹ may include the free distribution of ICS, sale to full cost or subsidized cost.

	through evidence	documentary	which the first ICS has been distributed under the VPA.	day when the first ICS is distributed to a household under this VPA.
			The start date of retroactive VPAs (with a start date prior to date of first submission of PoA) can be at the earliest 1 year prior to submission of documents for GS preliminary review.	The start date of the VPA will be checked by sales record, proof of shipping or any equivalent for the first ICS distributed under the VPA.
5	Methodology		Each VPA will comply with the applicability criteria of the applied methodology (TPDDTEC, version 03.1)	The VPA complies with all applicability criteria of TPDDTEC as further outlined in section B.2 of this document.
6	Financial Additionality & Ongoing Financial Needs		All projects (VPAs) to be included under the PoA will be in compliance with item 1.1.3 of Annex B – positive list mentioned in the 'Community Services Activity Requirements'. All VPAs will be solely composed of isolated units (efficient cookstoves) where the users of the technology/measure are household/SMEs/institutions and where each unit results in $\leq$ 1,800 MWh of thermal energy savings per year. Hence, according to paragraph 4.1.9 of the 'Community Services Activity Requirements', each of the VPAs, regardless of the host	The thermal energy savings per year at a unit level (i.e. per ICS) are clearly below 1,800 MWh as outlined in the ER calculation excel spreadsheet (worksheet 'Th. Energy savings unit level').

		country in which the project activity is being implemented, is deemed additional and therefore is not required to prove financial additionality at the time of Design Certification.
7	Stakeholder inclusivity	Local stakeholder consultation is done at VPA level, as described in section F of the PoA-DD. The Local stakeholder consultation report must be provided along with VPA-DD. A single Stakeholder consultation can be conducted for a group of VPAs as long as convincing justification is provided. A local stakeholder consultation report for a group of VPAs has been submitted to GS. The physical LSC meeting and stakeholder feedback round have been conducted (see LSC report for more details). The Local Stakeholder Consultation would be also valid for any other Voluntary Project Activities (VPAs) implemented in Kenya under BURN's Gold Standard PoA 'ECOABURN multi-country Clean Cooking Programme', provided that they are homogeneous, i.e. deploy the same stove type(s), target the same end-users and consist of the same project boundary as this specific VPA.
8	Conditions related to environmental Impact Analysis	The VPA has to fulfil host country requirements (if any) concerning environmental impact analysis. No EIA is required by the host country for ICS project activities as the

			one described in this document ²
9	CME approval	Each VPA has a project implementer that is either the Coordinating/Managing Entity or another entity that has signed a contractual agreement with the CME. Those agreements include all rights and responsibilities of both parties, e.g. approval procedures by the CME, monitoring requirements, carbon credit rights transfer. This eligibility criterion is not relevant if the CME is the VPA implementer.	This eligibility criterion is not relevant for this VPA since the CME is the same entity as the VPA implementer.
10	Transfer of carbon credit ownership	The transfer of carbon credit ownership all along the investment chain is clearly described and communicated to all project participants and end-users so that they are aware of to give up their rights on emission reductions. For technology producers and the retailers of the improved technology, this must be communicated by	The end-users permanently waive any claim or rights on carbon credits to the VPA implementer (at the same time CME of the PoA). This is confirmed by strap on ICS box and warranty booklet. There will be contractual agreements between distributors/retailers in which distributors/retailers

² See e.g. PoA9265, Appendix 8, https://cdm.unfccc.int/ProgrammeOfActivities/poa_db/BSVR8KUNAXW0LI1PGC6H42Y3JD9EQ7/view

		contract or clear written assertions in the transaction paperwork. The end-users will need to be informed and notified that they cannot claim for emission reductions from the project.	waive any claim or rights on carbon credits to the VPA implementer (at the same time CME of the PoA).The same will be submitted to the VVB during the validation.
11	Conditions to provide an affirmation that funding from Annex I Parties, if any, does not result in a diversion of ODA	In case that any of the VPA receives ODA, it is ensured that there is no diversion of ODA, i.e. that no ODA is provided under the condition that all or part of the carbon credits have to be returned to the donor country/entity providing ODA.	The VPA implementer has signed an ODA declaration confirming that there is no diversion of ODA. The same has been submitted to GS.
12	Target Group and distribution mechanism	The VPA serves households, institutions or SMEs either in urban, peri-urban and/or rural areas, and distributes the cook stoves through adequate distribution channels.	This VPA targets households in urban and peri-urban areas across the entire territory of Kenya. ICS are distributed through direct sale/distribution and/or a variety of retail outlets across the country to end-users.
13	Conditions related to sampling requirements	The VPA complies with the sampling plan as outlined VPA-DD, section B.7.2	The VPA-DD outlines the sampling plan in section B.7.2 which is in line with the one stipulated in the PoA-DD. The VPA will make part of a single sampling covering a group of VPAs, as soon as there are other homogenous VPAs. In case of a grouped sampling approach, the

			CDM Project Standard for PoAs will be followed.
14	Double counting of emission reductions	Each VPA will implement a unique identification system for every efficient cooking unit distributed to avoid double counting of emission reductions.	The unique identification system is explained in detail in section A.3. of this document. The VPA is in adherence to the CME Management System as outlined in Section C of the PoA-DD.
15	Crediting Period	The duration of the crediting period of the VPA does not exceed the end date of the registered PoA or shall be capped by the end date of the PoA. The final date for which ERs can be credited shall be no later than 20 years after the start date of the PoA.	The VPA will have a crediting period of 5 years which can be renewed twice, i.e. in total a maximum issuance of 15 years. The VPA will not exceed the end date of the registered PoA.

The VPA meets the requirements of the Community Services Activity Requirements, as follows –

Eligibility Criteria Category	Eligibility criterion - Required condition	Justification
1. Eligible Project Types	All CSA Projects shall lead to climate change mitigation and/or adaptation by providing or improving access to services/resources at the household or community or institution level. Eligible services include electricity and energy, water and sanitation, waste management, housing, etc.	The goal of the VPA is to distribute Improved Cook Stoves (ICS) in the households/SMEs of the host country of Kenya
GENERAL ELIGIBILITY CRITERIA		
2. Type of project	(b) End-use energy efficiency: Project activities that reduce energy requirements as compared to baseline scenario without affecting the level and quality of	The VPA involves distribution of energy efficient ICS.

Eligibility Criteria Category	Eligibility criterion - Required condition	Justification
	services or products, where the end-user of the products and services are clearly identified and when the physical intervention is required at the user end. For example, efficient cooking, heating, lighting, etc.	
3. Project Area, Boundary and scale	Project Area and Boundary shall be defined in line with the applicable Impact Quantification Methodologies and Product Requirements.	The project area is point location of CEP beneficiaries in the host country of the VPA. The project boundary will be limited to the geographical boundary of the host country of Republic of Kenya. For Improved Cookstoves, since TPDDTEC methodology is followed and there is no suppressed demand element, the guidelines of large-scale project shall be followed.
4. Legal Ownership	(a) Projects involving the distribution of a large number of devices for services such as heating, cooking, lighting, electricity generation, water treatment technology such as water filter, etc. shall provide a clear description of the ownership of the Products that are generated under Gold Standard Certification all along the investment chain. In line with the FPIC requirement, the proofs that end-users are aware of and willing to give up their rights on Products shall be provided. (b) The transfer of Product ownership shall be discussed	The end users confirm that rights to the ownership of carbon credits reside with the CME according to the end user agreement signed via monitoring app etc. (refer Eligibility under GS4GG section above).

Eligibility Criteria Category	Eligibility criterion - Required condition	Justification
	during local stakeholder consultations for projects.	

A.1.2. Legal ownership of products generated by the VPA and legal rights to alter use of resources required to service the project

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The project owner, BURN Manufacturing Co., has full and uncontested legal ownership of the Verified Emission Reductions (VERs) that are generated through the Gold Standard Certification.

BURN's ownership on the carbon credits has been clearly communicated to end-users and distributors.³

Further, as per Annex A of GHG Emissions Reduction and Sequestration Product Requirements version 2.1, as of now there are no mandatory caps enforced in the host country of Kenya for use of issued GS VERs for cookstove projects. However, if any such risk of double counting exists, the project developer commits to retiring eligible units equal to the quantity of Gold Standard VERs.

A.2. Location of VPA

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Republic of Kenya

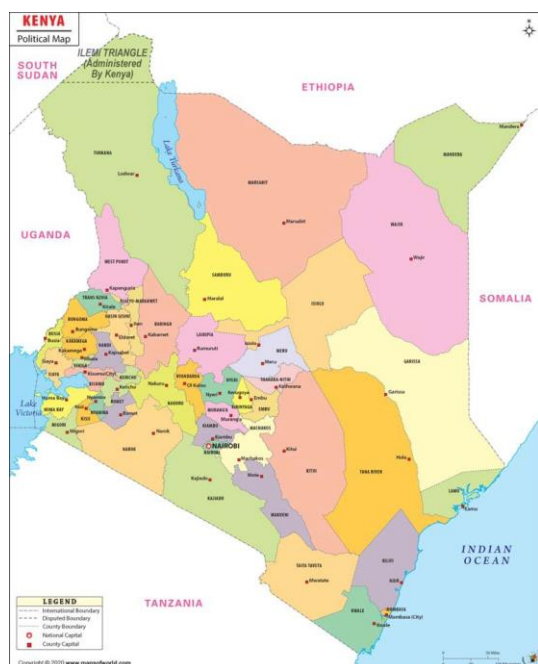
GPS coordinates⁴ for the Republic of Kenya:

Latitude: 0° 10' 36.73" N

Longitude: 37° 54' 29.98" E

³ The warranty booklet and a photo of the strip of the box have been uploaded to GS Registry.

⁴ <https://latitude.to/map/ke/kenya>



All the 47 Counties⁵ of the Republic of Kenya are covered by the VPA.

A.3. Technologies and/or measures

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The VPA deploys an efficient cookstove known as Jikokoa intended for use with charcoal. The technology was designed and developed by BURN. These highly efficient cookstoves translate into considerable charcoal savings when compared to traditional cookstoves. The Jikokoa stove's design takes into account the local cooking culture in the project area to ensure that improvements in technology and improved standards of living do not come at the expense of cultural traditions. The VPA implementer may opt to distribute other stove models in this VPA over time.

The households, by using the improved cookstoves (Jikokoa's), achieve considerable monetary savings through less charcoal consumption (positive contribution to SDG 1). Further, the households benefit of improved air quality. Since the Jikokoa's burn the fuel in a much cleaner way as the traditional stoves do (positive contribution to SDG 3). The project generates several jobs positively contributing to SDG 8.

⁵ https://en.wikipedia.org/wiki/Counties_of_Kenya

Please see the technical specifications in the following table:

Stove Manufacturer		BURN	
Stove Model		Jikokoa G3.5	
Stove Type		Charcoal Stove	
Materials			
Stove Body		CRCA Carbon Steel painted high gloss black epoxy powder coat	
Pot Rest		StainlessSteel	
Burning Chamber		StainlessSteel	
Ash Tray		Aluzinc	
Feet		StainlessSteel	
Measurements			
Height	cm	24.4 cm	
Diameter (stove top)	cm	26.0 cm	
Weight	kg	4 kg	
Fuel Chamber Volume	cm ³	954 cm ³	
Packaging Dimensions	cm	29.0 L x 28.5 W x 25.1 H	
WBT Results			
Parameter	Unit	Value	
High power thermal efficiency (average of cold start and hot start)	%	48.1%	
Firepower	kW	1.99	
Boil Time	minutes	27.59	
Lifetime			
Warranty	2 years		
Estimated Lifetime ³	7 - 10 years		

Stove Manufacturer		BURN	
Stove Model		Jikokoa Xtra (G4)	
Stove Type		Charcoal Stove	
Materials			
Stove Body		CRCA Carbon Steel painted high gloss black epoxy powder coat	
Pot Rest		StainlessSteel & Cast Iron	
Burning Chamber		StainlessSteel	
Ash Tray		Aluzinc	
Feet		Aluzinc	
Height	cm	27.0 cm	
Diameter (stove top)	cm	30.2 cm	
Weight	kg	5.5 kg	
Fuel Chamber Volume	cm ³	1030 cm ³	
Packaging Dimensions	cm	30.2 L x 30.5 W x 27.5 H	
WBT Results			
Parameter	Unit	Value	
High power thermal efficiency (average of cold start and hot start)	%	44.6%	
Firepower	kW	2.36	
Boil Time	minutes	26.44	
Lifetime			
Warranty	2 years		

Estimated Lifetime ⁶	7 to 10 years
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Stove Manufacturer	BURN	
Stove Model	Ecoa Char MMJ ⁷	
Stove Type	Charcoal Stove	
Materials		
Stove Body	CRCA Carbon Steel, painted hammertone black epoxy powder coat	
Pot Rest	Stainless Steel	
Burning Chamber	Stainless Steel	
Ash Tray	Aluzinc	
Feet	Aluzinc	
Handles	Stainless Steel and Polypropylene plastic ⁸	
Measurements		
Height	cm	22.8 cm
Diameter (stove top)	cm	26.7 cm
Weight	kg	3.0 kg
Fuel Chamber Volume	cm ³	1,152 cm ³
Packaging Dimensions	cm	29.5 L x 29.5 W x 24.0 H ⁹
WBT Results		
Parameter	Unit	Value

⁶ The lifetime of the Jikokoa Xtra may go beyond the indicated lifetime. Hence, depending on the usage rate of the stoves, stoves will be either removed from the database after the end of its lifetime and not credited anymore or remain in the database for crediting until the moment a significant drop in usage rate is observed. As an alternative, worn out ICS may be replaced by newly distributed stoves. Manufacturer's declaration about the ICS lifetime has been submitted to the VVB

⁷ The name of the stove model is still subject to change.

⁸ Not finally decided. Since stove model is in its final development phase.

⁹ Not finally decided. Since stove model is in its final development phase.

High power thermal efficiency (average of cold start and hot start)	%	49.29%
Firepower	kW	2.2
Boil Time	minutes	25.13
Lifetime		
Warranty	1 year	
Estimated Lifetime ¹⁰	7 - 10 years	



¹⁰The lifetime of the Ecoa Char MMJ may go beyond the indicated lifetime. Hence, depending on the usage rate of the stoves, stoves will be either removed from the database after the end of its lifetime and not credited anymore or remain in the database for crediting until the moment a significant drop in usage rate is observed. As an alternative, worn out ICS may be replaced by newly distributed stoves. Manufacturer's declaration about the ICS lifetime has been submitted to the validating DOE.



Photos: To the top left: Jikokoa Classic, To the top right: Jikokoa Xtra, To the bottom: ECOA Char MMJ.

Each ICS will be identified through an unique serial number (USN). The USN has the following format comprising of 9 digits¹¹:

1 st digit	2 nd digit	3 rd	4 th	5 th	6 th	7 th	8 th	9 th
Product ID	100000 th	10000 th	1000 th	100 th	10 th	Random	Random	1 st
ID	S1	S2	S3	S4	S5	R1	R2	S6

Each section on the USN will identify the product as follows:

- Product type: the first digit identifies the stove type (Jikokoa)
- # Production number: S1 to S6 are digit slots for a sequential numbering ordered by time of production, allowing for 1 million unique serial numbers. For instance, the first stove off the line would have "000000" for its S1-S6 digits.
- Random digits: R1 and R2 are 2 random digits placed in slots 7 & 8, to make the USN unpredictable to outside parties

Example for USN: 105097338

¹¹ It is possible that the USN format may change in future.

- "1" stands for Jikokoa product ID
- "050978" for S1-S6, meaning it was the 50,979th Jikokoa produced
- "33" for R1-R2, the random digits

The data for the system will be updated and modified as required to allow for optimal performance of VPA implementation and monitoring. All data will be stored for at least two (2) years after the expiry of the crediting period.

A.4. Scale of the VPA

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The VPA is a large-scale project activity. The applied TPDDTEC methodology allows large scale project activities. No suppressed demand baseline will be applied. The thermal energy savings per year at a unit level (i.e. per ICS) are clearly below 1,800 MWh as outlined in the ER calculation excel spreadsheet.

A.5. Funding sources of VPA

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The VPA is funded by private sources of the CME and possibly through private or public funds. In case that the VPA benefits of Official Development Assistance (ODA), it will be ensured that there is no diversion of ODA. See the signed ODA declaration uploaded to GS Registry.

SECTION B. APPLICATION OF APPROVED GOLD STANDARD METHODOLOGY (IES) AND/OR DEMONSTRATION OF SDG CONTRIBUTIONS

B.1. Reference of approved methodology (ies)

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Technologies and Practices to Displace Decentralized Thermal Energy Consumption, version 3.1.

B.2. Applicability of methodology (ies)

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Methodology applicability requirement	Justification regarding this VPA
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This methodology is applicable to programmes or activities introducing technologies and/or practices that reduce or displace greenhouse gas (GHG) emissions from the thermal energy consumption of households and non-domestic premises	This VPA introduces highly efficient improved charcoal cookstoves (ICS) reducing woody biomass consumption for urban and peri-urban households, hence reducing GHG emissions from the thermal energy consumption.
The project activity is implemented by a project proponent and can include additional project participants. The individual households and institutions do not act as project participants.	This VPA is implemented by BURN Manufacturing Co, at the same time being the CME of the PoA. The individual households do not act as project participants.
The project boundary needs to be clearly identified, and the technologies counted in the project are not included in any other voluntary market or CDM project activity (i.e. no double counting takes place). In some cases there maybe another similar activity within the same target area.	The geographical project boundary of this VPA is defined as the country of Kenya as described in section A.4 of this document. All carbon standard registries (UNFCCC, GS and VERRA) have been checked and it is confirmed that the VPA has not been registered as a separate GS project activity, nor included as part of another registered GS (or other carbon standard) PoA nor that the project activity has been deregistered. Hence, it can be confirmed that double counting is being avoided.
Project proponents must therefore have a survey mechanism in place together with appropriate mitigation measures so as to prevent any possibility of double counting.	The ICS under this VPA will avoid double accounting of emissions reductions through the Unique Serial Number (USN). Each device under this VPA is unquestionably assigned to the PoA 'ECO_A_BURN multi-country Clean Cooking Programme'. The USN will be clearly visible on the ICS throughout the life of the product as well as stored in the electronic data management system. If there is any doubt regarding the USN of a product it will be excluded from the VPA.

The technologies each have continuous useful energy outputs of less than 150kW per unit (defined as the total useful energy delivered from start to end of operation of a unit divided by time of operation). For technologies or practices that do not deliver thermal energy in the project scenario but only displace thermal energy supplied in the baseline scenario, the 150kW threshold applies to the displaced baseline technology.	The ICS promoted by this VPA have a capacity of less than the maximum 150kW per unit. The power output is between 1.9 to 2.4 kW (depending on the model).
Using the baseline technology as a backup or auxiliary technology in parallel with the improved technology introduced by the project activity is permitted as long as a mechanism is put into place to encourage the removal of the old technology (e.g. discounted price for the improved technology) and the definitive discontinuity of its use. The project documentation must provide a clear description of the approach chosen and the monitoring plan must allow for a good understanding of the extent to which the baseline technology is still in use after the introduction of the improved technology.	Customers will be encouraged to discontinue and remove the baseline technology. Nevertheless, some of the households may continue using the baseline stoves. The monitoring will provide information to what extent households continue using the baseline technologies after the introduction of the ICS: • Annual monitoring surveys will track the continued use of baseline technologies • The parameter, $P_{p,y}$, is based on subsumed Project Fuel Tests and thus use of traditional stove/fuel is accounted for in project emissions calculations.

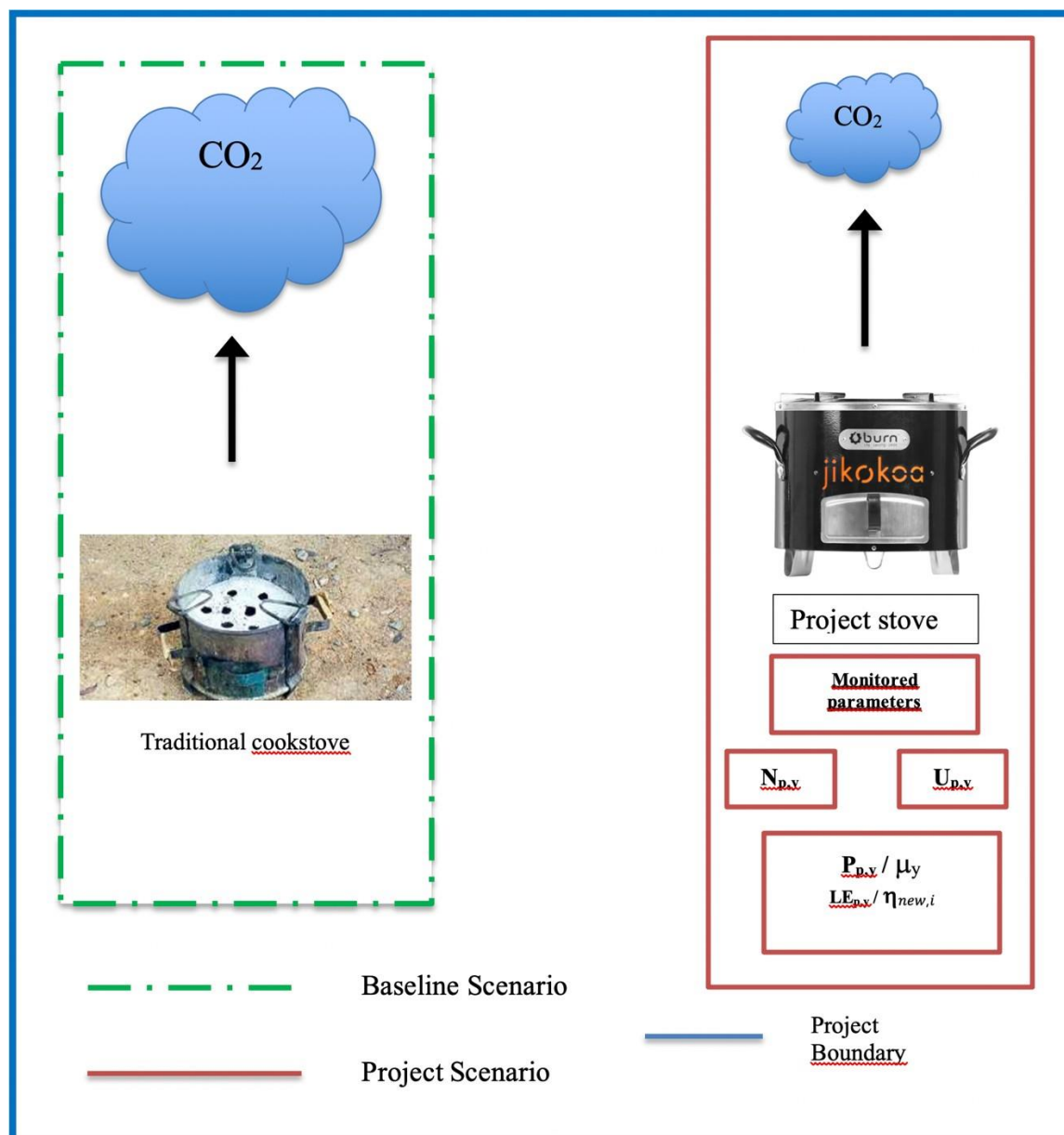
The project proponent must clearly communicate to all project participants the entity that is claiming ownership rights of and selling the emission reductions resulting from the project activity.	The project proponent will clearly communicate to all project participants that BURN Manufacturing Co. is claiming ownership rights and selling the ER credits resulting from the distribution of project technologies. • Transfer of carbon rights will be explained at local stakeholder consultations • Written messages on the stove packaging (strip on the box) and warranty booklet explaining that the rights on carbon credits are transferred from the end-users to BURN Manufacturing Co.
Project activities making use of a new biomass feedstock in the project situation (e.g. shift from non-renewable to green charcoal, plant oil or renewable biomass briquettes) must comply with relevant Gold Standard specific requirements for biomass related project activities, as defined in the latest version of the Gold Standard rules. If the biomass feedstock is sourced from a dedicated plantation, the criteria must apply to both plantations established for the project activity and existing plantations that were established in the context of other activities but will supply biomass feedstock.	This VPA does not involve use of a new biomass feedstock, hence this criterion is not applicable. End users continue to use non-renewable biomass in the project scenario.

Adequate evidence is supplied to demonstrate that indoor air pollution (IAP) conditions are not worsened compared to the baseline, and greenhouse gases (as listed in section II.1) emitted by the project fuel/stove combination are estimated with adequate precision. The project fuel/stove combination may include instances in which the project stove is a baseline stove.	This criterion is not applicable. Since this criterion is only relevant in case of introduction of a new biomass feedstock. Nevertheless, qualitative surveys are conducted as part of the monitoring surveys to investigate air quality with the project stove. The indoor air pollution is compared to the baseline scenario, asking end-users whether IAP increased, decreased or remained the same since the introduction of the ICS compared to the baseline stove.
Records of renewable fuel sales may not be used as sole parameters for emission reduction calculation, but may be used as data informing the equations in section II of this methodology. These records need to be correlated to data on distribution and results of field tests and surveys confirming (a) actual use of the renewable fuel and usage patterns (such as average fraction of non-renewable fuels used in mixed combustion or seasonal variation of fuel types), (b) GHG emissions, (c) evidence/justifications of CO levels not deteriorating (d) any further factors effecting emission reductions significantly.	This criterion is not applicable. Since this criterion is only relevant in case of introduction of a new biomass feedstock.

B.3. VPA boundary

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The project boundary is the physical/geographical site of the project technologies. Thus, the project boundary includes all individual households, which receive an ICS. The target area consists of households residing in urban and peri-urban areas across the 47 Counties of the Republic of Kenya. The fuel production and collection area is considered to be the same as the project boundary.



Source		GHGs	Included?	Justification/Explanation
Baseline scenario	Heat delivery	CO ₂	Yes	Important source of emission
		CH ₄	Yes	Important source of emissions
		N ₂ O	Yes	Significant for charcoal, hence included
Project scenario	Heat delivery	CO ₂	Yes	Important source of emission
		CH ₄	Yes	Important source of emissions

	N ₂ O	Yes	Significant for charcoal, hence included
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B.4. Establishment and description of baseline scenario

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Baseline assessment and KPT

As per the applied methodology TPDDTEC, the baseline scenario is defined by the typical baseline fuel consumption patterns in a population that is targeted for adoption of the project technology.

Biomass remains the predominant fuel for cooking in Kenya. Nationwide 54.6% and 14.6% of Kenya's household population utilize firewood and charcoal for cooking respectively thereby exerting enormous pressure on the environment. 84.3% of rural and 16.1% of urban population uses firewood for cooking. On the other hand, 8.9% of rural and 21.9 % of urban population uses charcoal for cooking¹²

The baseline fuel consumption has been determined from baseline KPT analysis and the resulting baseline charcoal consumption for the VPA is 0.6968t/household/year and baseline LPG consumption is 0.0031 t/HH-year. For more details, see section B.6.3 and ER calculation excel spreadsheet.

The suppressed demand baseline will not be used.

Calculation of Fraction of non-renewable biomass f_{NRB}

The f_{NRB} value has been taken from SeedEcology 3rd party report which has been calculated based on CDM Tool 30 EB 115 Annex 22 v4.0 2022.. The f_{NRB} value for Kenya is 0.9188

B.5. Demonstration of additionality

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¹² Table 3.18 Basic Report Based on 2015/16 Kenya Integrated Household Budget Survey

The table below is only applicable if the proposed project is deemed additional, as defined by the applied approved methodology or activity requirement or product requirement.

Specify the methodology, activity requirement or product requirement that establishes deemed additionality for the proposed project (including the version number and the specific paragraph, if applicable).	Community Services Activity Requirements (Version 1.2), paragraph 4.1.9: "Projects that meet any of the following criteria are considered as deemed additional and therefore are not required to prove Financial Additionality at the time of Design Certification: a) Positive list (Annex B) b) Projects located in LDC, SIDS, LLDC c) Microscale projects
Describe how the proposed VPA meets the criteria for deemed additionality.	The project falls under the positive list (Annex B, item 1.1.3). The project is composed of isolated units where the users of the technology are households and where each unit results in $\leq 1,800$ MWh (i.e. 9.47 MWh/stove/year) thermal energy savings per year (see worksheet tab 'Th.Energy savings unit level' of the ER calculation excel spreadsheet). Hence paragraph 4.1.9, (a) as per the Community Services Activity Requirements is met.

B.5.1. Prior Consideration

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The time of first submission (1/11/2021) is within one year of the project start date (1/12/2020). This can be double checked on the Sustain-Cert registry.

B.5.2. Ongoing Financial Need

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NA

B.6. Sustainable Development Goals (SDG) outcomes

Sustainable Development Goals Targeted	Most relevant SDG Target	SDG Impact
		Indicator (Proposed or SDG Indicator)
13 Climate Action (mandatory)	Target 13.3	Emission Reductions
1 End poverty in all its forms everywhere	Target 1.4	Monetary savings related to the purchase of charcoal
3 Ensure healthy lives and promote well-being for all at all ages	Target 3.9	Perceived airquality
7 Ensure access to affordable, reliable sustainable and modern energy for all	Target 7.1	Number of sold/distributed ICS in use
8 Promote sustained, inclusive and sustainable economic growth, full and productive employment and decent workfor all	Target 8.5	Number of jobs created

B.6.1. Explanation of methodological choices/approaches for estimating the SDG Impact

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The outcome of the SDG 13 will be measured as reduced CO₂e emissions applying the GS methodology TPDDTEC. The SDG 13 outcome will be certified as "Certified SDG 13 Impacts" allowing the generation of carbon credits (GS VERs).

This VPA distributes highly efficient charcoal stoves which reduce charcoal consumption. The baseline fuel and the project fuel are the same and the baseline emission factor and project emission factor are considered the same. The overall GHG reductions achieved by the project activity in year y are calculated as follows:

$$ER_y = \sum_{b,p} (N_{p,y} * U_{p,y} * P_{p,b,y} * NCV_{b,fuel} * (f_{NRB,b,y} * EF_{fuel,CO2} + EF_{fuel,nonCO2})) - \sum LE_{p,y}$$

Equation (1)

Where:

ER_y	3.32	Emission reduction per stove per year (tCO ₂ e/year)
$N_{p,y}$	365	Cumulative number of project technology-days included in the project database for project scenario p against baseline scenario b in year y
$U_{p,y}$	90%	Cumulative usage rate for technologies in project scenario p in year y, based on cumulative adoption rate and drop off rate revealed by usage surveys (fraction)
$P_{p,b,y}$	2.19	Specific fuel savings for an individual technology of project p against an individual technology of baseline b in year y, in tons/year, as derived from the statistical analysis of the data collected from the field tests
$f_{NRB,b,y}$	0.9188	Fraction of biomass used in year y for baseline scenario b that can be established as non-renewable biomass
$NCV_{b,fuel}$	0.0156	Net calorific value of the fuel that is substituted or reduced. (IPCC default for wood fuel, 0.0156 TJ/ton)
$EF_{b,fuel,CO2}$	112	CO ₂ emission factor of the fuel that is reduced. (IPCC default for charcoal fuel, 112 tCO ₂ /TJ)
$EF_{b,fuel,nonCO2}$	9.46	Non-CO ₂ emission factor of the fuel that is reduced
$LE_{p,y}$	0.00	Leakage for project scenario p in year y (tCO ₂ e/yr)

Baseline emissions (LPG use)	0.0083	Baseline emissions from LPG use in year y (tCO ₂ e/yr)
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Project emissions (LPG use)	0.1348	Project emissions from LPG use in year y (tCO ₂ e/yr)
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Net Emission reductions after adjusting emissions from LPG use= $ER_y + \text{Baseline emissions (LPG use)} - \text{Project emissions (LPG use)}$

$$= 3.45 + 0.0083 - 0.1348$$

$$= 3.32 \text{ tCO}_2/\text{yr}$$

Methodological choices/approaches related to SDG 1

The contribution of the project to SDG 1 will be confirmed by a random sample survey with a representative number of households in which the money spent for charcoal for preparing meals in the project scenario will be compared to the baseline scenario. It will be checked on whether households achieve indeed monetary savings using the improved charcoal cookstove, which would provide evidence that the project positively contributes to SDG 1.

Methodological choices/approaches related to SDG 3

The contribution of the project to SDG 3 will be confirmed by a random sample survey with a representative number of households in which pollution-related inconveniences (such as smoke levels, itchy eyes and breathing problems) in the project scenario will be compared to the baseline scenario. In case that households confirm that due to the project there is less pollution-related inconveniences compared to the baseline scenario, it provides evidence that the project positively contributes to SDG 3.

Methodological choices/approaches related to SDG 7

The contribution of the project to SDG 7 will be confirmed by the number of distributed improved charcoal stoves which are in use.

This is reflected by the the following: $N_{p,y} * U_{p,y}$

Methodological choices/approaches related to SDG 8

The contribution of the project to SDG 8 will be confirmed by the number of jobs created.

B.6.2. Data and parameters fixed ex ante

SDG13

Data/parameter	EF _{b,CO2}
Unit	tCO ₂ /TJ
Description	CO2 emission factor arising from use of fuel wood in baseline scenario
Source of data	2006 IPCC Guidelines for National Greenhouse Gas Inventories, volume 2, chapter 2 (Table 2.5)
Value(s) applied	112
Choice of data or Measurement methods and procedures	Default IPCC value for fuel wood is applied
Purpose of data	Calculation of baseline emissions
Additional comment	-

Data/parameter	EF _{b,non-CO2}				
Unit	tCO ₂ e/TJ				
Description	Non-CO2 emission factor arising from use of fuel wood in baseline scenario				
Source of data	2006 IPCC Guidelines for National Greenhouse Gas Inventories, volume 2, chapter 2 (Table 2.5)				
Value(s) applied	9.46				
Choice of data or Measurement	Table 2.5, volume 2, chapter 2, 2006 IPCC Guidelines for National Greenhouse Gas Inventories				

methods and procedures					
Purpose of data	Non-CO2 Emission calculation in baseline				
Additional comment	-				

Data/parameter	EF_{p,CO_2}
Unit	tCO ₂ /TJ
Description	CO ₂ emission factor arising from use of fuel wood in project scenario
Source of data	2006 IPCC Guidelines for National Greenhouse Gas Inventories, volume 2, chapter 2 (Table 2.5)
Value(s) applied	112
Choice of data or Measurement methods and procedures	Default IPCC value for fuel wood is applied.
Purpose of data	CO ₂ emission calculation in project scenario
Additional comment	-

Data/parameter	$EF_{p,non-CO_2}$
Unit	tCO ₂ e/TJ

Description	Non-CO ₂ emission factor arising from use of fuel wood in project scenario
Source of data	2006 IPCC Guidelines for National Greenhouse Gas Inventories, volume 2, chapter 2 (Table 2.5)
Value(s) applied	9.46
Choice of data or Measurement methods and procedures	Table 2.5, volume 2, chapter 2, 2006 IPCC Guidelines for National Greenhouse Gas Inventories
Purpose of data	Non-CO ₂ emission calculation in project scenario
Additional comment	-

Data/parameter	NCV _b
Unit	TJ/ton of fuel wood
Description	Net calorific value of the fuel wood used in baseline
Source of data	IPCC default 2006, volume 2, chapter 1 (Table 1.2)
Value(s) applied	0.0156
Choice of data or Measurement methods and procedures	Default IPCC value for fuel wood is applied.
Purpose of data	CO ₂ emission calculation in baseline scenario
Additional comment	-

Data/parameter	NCV_p
Unit	TJ/ton of fuel wood
Description	Net calorific value of the fuel wood used in project scenario
Source of data	IPCC default 2006, volume 2, chapter 1 (Table 1.2)
Value(s) applied	0.0156
Choice of data or Measurement methods and procedures	Default IPCC value for fuel wood is applied.
Purpose of data	CO ₂ emission calculation in project scenario
Additional comment	-

Data/parameter	$f_{NRB,y}$
Unit	fraction
Description	Non-renewability status of woody biomass fuel in scenario i during year y
Source of data	SeedEcology 3rd party report
Value(s) applied	0.9188
Choice of data or Measurement methods and procedures	f_{NRB} assessment based on CDM Tool 30 EB 115 Annex 22 v4.0 2022
Purpose of data	ER calculation
Additional comment	The f_{NRB} value will remain fixed during the crediting period

Data/parameter	Wood-to-charcoal conversion factor
Unit	kg firewood / kg charcoal
Description	Conversion factor for transforming fuel wood into charcoal
Source of data	IPCC default value
Value(s) applied	6
Choice of data or Measurement methods and procedures	Default IPCC value (1996 IPCC Guidelines for National Greenhouse Gas Inventories) is applied. (https://www.ipcc-nggip.iges.or.jp/public/gl/guidelin/ch1ref3.pdf)
Purpose of data	Used to calculate fuel savings in fuel wood equivalent
Additional comment	Fixed ex-ante at VPA level

Data/parameter	$P_{b,y}$
Unit	t/household-day
Description	Quantity of woody biomass that is consumed in baseline scenario b during year y
Source of data	Baseline KPT
Value(s) applied	0.01 (4.18 t/household-year)
Choice of data or Measurement methods and procedures	This value is based on baseline KPT (Baseline survey results provided to VVB for more details)
Purpose of data	Used to calculate the fuel savings

Additional comment	The baseline will remain by-default fixed during the crediting period since the project activity targets non-industrial applications (see page 6 of TPDDTEC)
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Data/parameter	$P_{b,LPG,y}$
Unit	t/household-day
Description	Quantity of LPG that is consumed in baseline scenario b during year y
Source of data	Baseline KPT
Value(s) applied	0.00009
Choice of data or Measurement methods and procedures	This value is based on baseline KPT (Baseline survey results provided to VVB for more details)
Purpose of data	Used to calculate the Emission Reductions
Additional comment	The baseline will remain by-default fixed during the crediting period since the project activity targets non-industrial applications (see page 6 of TPDDTEC)

Data/parameter	NCV_{LPG}
Unit	TJ/ton
Description	Net calorific value of the LPG
Source of data	IPCC default 2006, volume 2, chapter 1 (Table 1.2)
Value(s) applied	0.0473
Choice of data or Measurement	Default IPCC value for LPG is applied

methods and procedures	
Purpose of data	Calculation of baseline and project emissions
Additional comment	-

Data/parameter	EF _{LPG,CO2}
Unit	tCO ₂ /TJ
Description	CO ₂ emission factor arising from use of LPG
Source of data	2006 IPCC Guidelines for National Greenhouse Gas Inventories, volume 2, chapter 2 (Table 2.2)
Value(s) applied	63.1
Choice of data or Measurement methods and procedures	Default IPCC value for LPG is applied
Purpose of data	CO ₂ emission calculation
Additional comment	-

B.6.3. Ex ante estimation of SDG Impact

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Ex-ante calculations related to the outcomes of SDG 13

The transparent ex-ante calculations of the outcomes of SDG 13 (i.e. CO₂e reductions) are provided in a separate excel spreadsheet uploaded to GS registry. For data/parameters available before design certification, values contained in section

B.6.3 and for data/parameters not available before design certification and monitored during the crediting period, the estimates contained in section B.7.1 have been used.

Ex-ante calculations related to the outcomes of SDG 1

N/A. For ex-ante purposes, it is estimated that households achieve monetary savings of around 50.89% in average. The monitoring of SDG 1 is made through a qualitative evaluation of sample households during the usage/monitoring survey. For this aspect no specific calculations are needed.

Ex-ante calculations related to the outcomes of SDG 3

N/A. For ex-ante purposes, it is estimated that at least 98.76% of households perceive improved air quality. Since monitoring of SDG 3 is made through a qualitative evaluation of sample households during the usage/monitoring survey. For this aspect no specific calculations are needed.

Ex-ante calculations related to the outcomes of SDG 7

The parameter 'project technologies in use' will be calculated as part of the outcome calculation for SDG 13 and is provided in the separate ER calculation excel spreadsheet. The distributed cookstoves ($N_{p,y}$) are multiplied with the usage rate ($U_{p,y}$) to determine the 'project technologies in use'.

Ex-ante calculations related to the outcomes of SDG 8

N/A. For ex-ante purposes, the number of created jobs has been estimated to be 25 and no specific calculation is needed.

Monitoring of SDG 8 is conducted through the employee lists summing up the total number of jobs created.

B.6.4. Summary of ex ante estimates of each SDG outcome

SDG 13

The use of efficient cookstoves in project activity would result in reduction in charcoal consumption which leads to lesser CO₂e emissions. The project would result in average 171,888 tCO₂e emission reductions over the crediting period of 5 years

Year	Baseline estimate	Project estimate	Net benefit
Year 1	65,057	31,057	34,000
Year 2	196,979	94,035	102,944
Year 2	328,901	157,013	171,888
Year 4	460,823	219,990	240,833
Year 5	592,745	282,968	309,777
Total	1,644,505	785,063	859,442
Total number of crediting years	5		
Annual average over the crediting period	328,901	157,013	171,888

SDG 1

Less money is spent for purchasing charcoal in the project scenario compared to the baseline scenario. It is estimated that households achieve monetary savings of around 50.89% in average.

Year	Baseline estimate	Project estimate	Net benefit
Year 1	0%	50.89% ¹³	50.89%
Year 2	0%	50.89%	50.89%
Year 2	0%	50.89%	50.89%
Year 4	0%	50.89%	50.89%
Year 5	0%	50.89%	50.89%
Total	NA	NA	NA

¹³ In the project activity, households save 50.89% of charcoal instead of 0% in the baseline scenario. This results in charcoal savings of 50.89% (indicated in the column 'Net benefit').

Total number of crediting years	5		
Annual average over the crediting period	0%	53.08%	53.08%

SDG 3

Less pollution-related inconveniences (such as smoke levels, itchy eyes and breathing problems) in the project scenario compared to the baseline scenario. It is estimated that at least 98.76%98.76%98.76%% of households perceive improved air quality.

Year	Baseline estimate	Project estimate	Net benefit
Year 1	0%	98.76% ¹⁴	98.76%
Year 2	0%	98.76%	98.76%
Year 2	0%	98.76%	98.76%
Year 4	0%	98.76%	98.76%
Year 5	0%	98.76%	98.76%
Total	NA	NA	NA
Total number of crediting years	5		
Annual average over the crediting period	0%	98.76%	98.76%

¹⁴ The value is based on the Usage survey carried out for the first Verification of the project activity

SDG 7

It is estimated that 44,877 ICS will have been distributed and be in use by the end of the first crediting period. An average usage rate of 90% is expected.

Year	Baseline estimate	Project estimate	Net benefit
Year 1	0	8,877	8,877
Year 2	0	26,877	26,877
Year 2	0	44,877	44,877
Year 4	0	62,877	62,877
Year 5	0	80,877	80,877
Total	NA	NA	NA
Total number of crediting years	5		
Annual average over the crediting period	0	44,877	44,877

SDG 8

It is estimated that the project creates at least 25 jobs over 5 years (1st crediting period).

Year	Baseline estimate	Project estimate	Net benefit
Year 1	0	25	25
Year 2	0	25	25
Year 2	0	25	25
Year 4	0	25	25
Year 5	0	25	25
Total	0	25	25

Total number of crediting years	5		
Annual average over the crediting period	0	25	25

B.7. Monitoring plan

B.7.1. Data and parameters to be monitored

SDG 13

Data / Parameter	$N_{p,y}$
Unit	Number of project cookstove credited (units)
Description	Cookstoves in the project database for project scenario p through year y
Source of data	Distribution database
Value(s) applied	For ex-ante estimation, it has been assumed that 49,863 stoves are being distributed. ¹⁵ Since cookstove sales are progressive, it is assumed that cookstoves are in operation (thus credited) for 180 days in average in the year when they are sold.
Measurement methods and procedures	BURN keeps records of all distributed ICS in an electronic database. As a minimum the following information will be recorded through EchoMobile16 in the database: • Unique serial number (USN) of the ICS • Date of shipment to distributor/retailer • Name of distributor/retailer • Quantity of ICS distributed • Geographic area of distributor/retailer • Model type of the ICS

¹⁵ This number is a preliminary estimate and the actual number ex-post can be lower or higher.

¹⁶ <https://www.echomobile.org>

	Besides, the distribution database will contain end-user contact details (name, address, mobile number, or national ID number, mode of use) of at least 10 times the survey and field test sample size (including usage surveys for each age of product), in order to ensure an adequate end-user pool to which random sampling can be applied. In order to claim the ICS warranty, end-users have to register their end-user details through SMS or call.
Monitoring frequency	Continuously
QA/QC procedures	Transparent data analysis and reporting
Purpose of data	Estimation of CO ₂ e emission reductions
Additional comment	The total distribution record is divided based on project scenario to create the project database.

Data / Parameter	$U_{p,y}$
Unit	Percentage
Description	Usage rate in project scenario p during year y
Source of data	Assumption
Value(s) applied	90% (for ex-ante estimation)
Measurement methods and procedures	The usage survey is carried out annually as described in section B.7.2 of this VPA-DD
Monitoring frequency	Annual
QA/QC procedures	Transparent data analysis and reporting
Purpose of data	Estimation of CO ₂ e emission reductions
Additional comment	A single usage parameter is weighted to be representative of the quantity of project technologies of each age being credited in a given project scenario. The Good Practice level as per the 'GS Requirements and Guidelines: Usage rate Monitoring' is being followed. The VPA may follow the best practice later during the crediting period provided that a continuous use monitoring can be implemented.

	A user is defined as a household which uses the project stove for at least 7 meals per week. Usage will be confirmed during monitoring/usage surveys.
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Data / Parameter	$P_{p,y}$
Unit	t/household-year
Description	Quantity of woody biomass that is consumed in project scenario p during year y
Source of data	Project KPT
Value(s) applied	0.33t/household-year
Measurement methods and procedures	Determined through project KPT. The procedure as per Annex 4 of the applied methodology TPDDTEC will be followed.
Monitoring frequency	Every 2 years or in case that aging test approach (as per Annex 8 of the applied methodology) is applied, once prior to 1 st issuance.
QA/QC procedures	Transparent data analysis and reporting
Purpose of data	Used to calculate the fuel savings
Additional comment	A single project fuel consumption parameter is weighted to be representative of the quantity of project technologies of each age being credited in a given project scenario.

Data / Parameter	$P_{p,LPG,y}$
Unit	t/household-day
Description	Quantity of LPG that is consumed in project scenario p during year y
Source of data	Project KPT
Value(s) applied	0.00014

Measurement methods and procedures	Determined through project KPT. The procedure as per Annex 4 of the applied methodology TPDDTEC will be followed.
Monitoring frequency	Every 2 years
QA/QC procedures	Transparent data analysis and reporting
Purpose of data	Used to calculate the emission reductions
Additional comment	-

Data / Parameter	$LE_{p,y}$
Unit	tCO ₂ e per year
Description	Leakage in project scenario p during year y
Source of data	Baseline and monitoring surveys
Value(s) applied	0 (for ex-ante estimation)
Measurement methods and procedures	-
Monitoring frequency	Every two years
QA/QC procedures	Transparent data analysis and reporting
Purpose of data	Used to calculate leakage emissions
Additional comment	Aggregate leakage can be assessed for multiple project scenarios, if appropriate

Data / Parameter	$\eta_{new,i}$
Unit	Fraction
Description	Efficiency of the ICS of each type i being deployed as part of the project activity

Source of data	Water Boiling Tests (WBT) following the WBT protocol, 4.2.3. (https://www.cleancookingalliance.org/technology-and-fuels/testing/protocols.html)
Value(s) applied	Monitored parameter
Measurement methods and procedures	The procedures as outlined in Annex 8 of TPDDTEC will be followed. The minimum sample size of each age group shall comply with the 90/10 rule.
Monitoring frequency	Annual
QA/QC procedures	-
Purpose of data	To calculate project fuel consumption
Additional comment	Only applicable if the ageing test approach (as described in Annex 8 of TPDDTEC) is chosen instead of the biennial project KPTs to account for changes in the project scenario over time as project technologies age.

Data / Parameter	μ_y
Unit	Fraction
Description	Adjustment to account for any continued use of pre-project devices (baseline stove) in the project scenario during the year y
Source of data	Monitoring/usage surveys
Value(s) applied	Monitored parameter
Measurement methods and procedures	Any of the following approach with justification on how the chosen option fits within project circumstances Measurement campaigns shall be undertaken using data loggers such as stove utilization monitors (SUMs) which can log the operation of all devices in order to determine the average device utilization intensity

	Monitoring surveys to capture cooking habits and stove usage of households in the region, including quantification of use of baseline devices, by formulating questions and/or collecting evidences to determine the frequency of usage of both the project devices and baseline devices. Monitoring surveys to capture the number of meals cooked
Monitoring frequency	Annual
QA/QC procedures	-
Purpose of data	Emission reduction calculation
Additional comment	Only applicable if the ageing test approach (as described in Annex 8 of TPDDTEC) is chosen instead of the biennial project KPTs to account for changes in the project scenario over time as project technologies age. No need to monitor this parameter in case that biennial project KPTs are conducted, since in this case the KPTs capture the total project fuel consumption on all stoves, i.e. also includes the fuels consumed on any baseline stoves.

SDG 1

Data / Parameter	Monetary savings related to the purchase of charcoal
Unit	-
Description	Monetary savings related to the purchase of charcoal
Source of data	Survey
Value(s) applied	50.89% (for ex-ante estimate)
Measurement methods and procedures	Carrying out surveys (either site visits or telephone surveys) to check on the money spent for purchasing

	charcoal in the project scenario compared to the baseline scenario.
Monitoring frequency	At least once every two years (biennial)
QA/QC procedures	-
Purpose of data	Reporting on SDG 1
Additional comment	This parameter is measured qualitatively, but not quantitatively.

SDG 3

Data / Parameter	Perceived air quality
Unit	-
Description	Smoke levels, itchy eyes and breathing problems
Source of data	Survey
Value(s) applied	98.76% perceive an improved air quality (ex-ante estimate)
Measurement methods and procedures	Carrying out surveys (either site visits or telephone surveys) to check on the pollution-related inconveniences (such as smoke levels, itchy eyes and breathing problems) in the project scenario compared to the baseline scenario.
Monitoring frequency	At least once every two years (biennial)
QA/QC procedures	-
Purpose of data	Reporting on SDG 3
Additional comment	This parameter is measured qualitatively, but not quantitatively.

SDG 7

Data / Parameter	Number of sold/distributed ICS in use
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Unit	Number of units in use
Description	Number of sold/distributed ICS in use
Source of data	Project database
Value(s) applied	44,877 ICS are estimated to be distributed (annual average). The usage rate is expected to be 90%.
Measurement methods and procedures	The total number of ICS sold/distributed is summed up in the database
Monitoring frequency	Continuously
QA/QC procedures	-
Purpose of data	Reporting on SDG 7
Additional comment	-

SDG 8

Data / Parameter	Number of jobs created
Unit	Number
Description	Number of jobs created
Source of data	Project records like contracts, payment slips, employee list or others
Value(s) applied	25 jobs expected to be created
Measurement methods and procedures	-
Monitoring frequency	Annually
QA/QC procedures	Employee list can be cross-checked with contracts/payment slips or others
Purpose of data	Reporting on SDG 8
Additional comment	-

B.7.2. Sampling plan

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Ongoing monitoring

Usage survey: An annual usage survey determines the drop off rates as project technologies age and users switch back to the baseline technology¹⁷. The usage parameter will be weighted to be representative of the quantity of project technologies of each age being credited in a given project scenario. The minimum total sample size is 100 randomly selected households, with at least 30 samples for project technologies of each age being credited. The majority of interviews will be conducted in person by BURN staff or by hired externals which would be trained before and include expert observation by the interviewer within the kitchen in question, while the remainder may be conducted via telephone by the same interviewers on condition that in-kitchen observational interviews are first concluded and analyzed such that typical circumstances are well understood by the telephone interviewers.

Monitoring survey: Along with the usage survey, a monitoring survey is carried out annually to assess end-user characteristics such as technology use, fuel consumption and seasonal variation.

At least with every 2nd monitoring survey a leakage assessment will be conducted. The leakage assessment evaluates if the project has in any way lead to an increase in emissions outside of the project boundary and if the increase will be directly attributed to the project activity. The leakage assessment is done in the following:

Potential leakage source	Probability	Justification
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¹⁷ It may be the case that the drop off rate is lower in the second year than in the first year, reflecting possible difficulties in the early adoption of a new technology.

The displaced baseline technologies are reused outside the project boundary in place of lower emitting technology or in a manner suggesting more usage than would have occurred in the absence of the project.	Very low	The displaced baseline technology is the most common and easily available cooking method in the project area and most of households usually sale off the old stove as scrap metal or throw it away upon purchase of ICS. It is highly unlikely that displaced baseline technology is reused outside the project boundary.
Non-project users who previously used lower emitting energy sources use the non-renewable biomass or fossil fuels saved under the project activity.	Can be excluded	Project users have to spend money for the charcoal. It can be excluded that the fuel saved by the project would be given for free by the project users and used by non-project users who previously used lower emitting energy sources.
The project significantly impacts the NRB fraction within an area where other CDM or VER project activities account for NRB fraction in their baseline scenario.	Very low	The project is too small that it would significantly have an impact on the NRB fraction. Besides, demand for charcoal in Kenya is continuously rising. Since alternative fuels (like LPG or electricity) are out of reach for the majority of people. Hence, the share of NRB remains high and it will not have a leakage impact on other carbon projects in Kenya.
The project population compensates for loss of the space heating effect of inefficient technology by adopting some other form of heating or by retaining some use of inefficient technology.	Very low	The climate conditions for most of the areas in Kenya do usually not require space or room heating. It is very unlikely that the charcoal ICS would be used for space or room heating. This can be confirmed through annual monitoring surveys.

By virtue of promotion and marketing of a new technology with high efficiency, the project stimulates substitution within households who commonly used a technology with relatively lower emissions, in cases where such a trend is not eligible as an evolving baseline.	Can be excluded	The project's target group is households using charcoal. It is highly unlikely that households using LPG or electricity for cooking would use the project technology. Thus, leakage can be excluded.
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The PP will update the project fuel consumption by carrying out biennial *project KPTs* to account for changes in the project scenario over time as project technologies age. Alternatively, the PP will monitor following Annex 8 of TPDDTEC the degradation in the performance of cookstove efficiency following the WBT and accordingly adjust the project fuel consumption level.

In case of choosing the first option, KPTs would follow the 90/30 rule in case of a paired or independent sampling or the 90/10 rule in case of single sample tests. In case that the 90/30 rule (in case of paired or independent sampling) or 90/10 rule (in case of single sample test) is not met, additional random samples will be taken or the upper bound of the 90% confidence interval will be applied. The procedures as outlined in section 7 and Annex 4 of the applied methodology will be followed.

In case PP opts for the *ageing test approach* instead of biennial project KPTs, annual WBTs would be conducted on a representative sample of each age group. The sample size of each age group will be big enough so that the results comply with the 90/10 rule. The WBT shall be carried out along with the project KPTs prior to 1st issuance and then subsequently WBTs shall be carried out annually to monitor the degradation in the efficiency of the ICS. The WBTs should be conducted in the last 3 months of the monitoring period, provided it is representative of annual conditions. Choosing the ageing test approach, the PP would have to ensure to a) raise additional questions in the monitoring survey related to the frequency of usage of both the project and baseline devices or capture the number of meals cooked or b) carry out measurement campaigns using data loggers such as stove utilization monitors (SUMs) to take into account for the parallel use of baseline stoves in the project scenario or c) capture the number of

meals cooked on both project and baseline devices. The measurement campaign would comprise of at least 100 randomly selected households for at least 90 days, with at least 30 samples for project technologies of each age being credited. during the year y .

All of the aforementioned surveys/tests (except WBTs) are intended to be carried out with smartphones (by using e.g. Echomobile software tool) and as soon as the user is online all the captured data are transferred to a centralised database. The team of BURN and the external carbon consultant carries out from there the further data analysis. If smartphone technology for any reason cannot be used, data will be recorded in paper form.

The sampling frame for the aforementioned ICS surveys comprises of all households making part of the carbon database.

The sampling frame for the KPTs/WBTs consist of all households using the project ICS.

The surveys/tests will be carried out by BURN staff or by hired externals which would be trained before. Quality of the data will be checked by the carbon consultant.

The survey for SDGs 1 and 3 follows the same sampling design as the usage survey to determine the usage rate. The households to be surveyed for SDGs 1 and 3 will be the same as the ones surveyed for $U_{p,y}$.

The VPA will make part of a single sampling covering a group of VPAs, as soon as there are other homogenous VPAs, i.e. apply the same ICS model (Jikokoa stove) and fuel (charcoal), and target the same end-users (i.e. households) within the same geographical area, i.e. Republic of Kenya. In case of a grouped sampling approach, the CDM Project Standard for PoAs will be followed.

B.7.3. Other elements of monitoring plan

>>

Monitoring of $N_{p,y}$

BURN keeps records of all distributed ICS in an electronic database. As a minimum the following information will be recorded through EchoMobile¹⁸ in the database:

- Unique serial number (USN) of the ICS
- Date of shipment to distributor/retailer
- Name of distributor/retailer
- Quantity of ICS distributed
- Geographic area of distributor/retailer
- Model type of the ICS

Besides, the distribution database will contain end-user contact details (name, address, mobile number, or national ID number, mode of use) of at least 10 times the survey and field test sample size (including usage surveys for each age of product), in order to ensure an adequate end-user pool to which random sampling can be applied. In order to claim the ICS warranty, end-users have to register their end-user details through SMS or call.

The USN of each ICS entered into the distribution database will be linked to a distribution date (recorded during distribution). Thus, for any monitoring period it is possible to calculate the period of time for which the stoves included in the emissions reduction calculations are deemed operating. If e.g. a stove has been operating for 180 days, then the full-year operating fraction is 0.493 (=180/365 days). A stove will be counted as operational (= start crediting) from the next day following the stove distribution. The sum of the operating fractions of all appliances determines the equivalent full-time appliances for the monitoring period.

The USN has the following format comprising of 9 digits¹⁹:

1 st digit	2 nd digit	3 rd	4 th	5 th	6 th	7 th	8 th	9 th
Product ID	100000 th	10000 th	1000 th	100 th	10 th	Random	Random	1 st
ID	S1	S2	S3	S4	S5	R1	R2	S6

¹⁸ <https://www.echomobile.org>

¹⁹ It is possible that the USN format may change in future.

Each section on the USN will identify the product as follows:

- Product type: the first digit identifies the stove type (e.g. Kuniokoa)
- # Production number: S1 to S6 are digit slots for a sequential numbering ordered by time of production, allowing for 1 million unique serial numbers. For instance, the first stove off the line would have "000000" for its S1-S6 digits.
- Random digits: R1 and R2 are 2 random digits placed in slots 7 & 8, to make the USN unpredictable to outside parties

Example for USN: 202728110

- "2" stands for Kuniokoa product ID
- "027280" for S1-S6, meaning it was the 27,281st Kuniokoa produced
- "11" for R1-R2, the random digits

The data for the system will be updated and modified as required to allow for optimal performance of each VPA implementation and monitoring. All data will be stored for at least two (2) years after the expiry of the crediting period.

SECTION C. DURATION AND CREDITING PERIOD

C.1. Duration of project

C.1.1. Start date of VPA

>>01/12/2020 (This is date of 1st ICS sales)

C.1.2. Expected operational lifetime of VPA

>>15 years (3 x 5 years)

C.2. Crediting period of project

C.2.1. Start date of crediting period

>>01/01/2021

C.2.2. Total length of crediting period

>>5 years. The crediting period may be renewed twice in line with the Community Services Activity Requirements.

SECTION D. SUMMARY OF SAFEGUARDING PRINCIPLES AND GENDER SENSITIVE ASSESSMENT

D.1. Safeguarding Principles that will be monitored

A completed Safeguarding Principles Assessment is in [Appendix 1](#), no ongoing monitoring is applicable to this VPA.

D.2. Assessment that project complies with GS4GG Gender Sensitive requirements

Question 1 - Explain how the project reflects the key issues and requirements of Gender Sensitive design and implementation as outlined in the Gender Policy?	The project activity doesn't endorse any form of discrimination based on gender. Kenya has ratified ILO Conventions 45 (Underground Work (Women) and ILO Convention 111 (Discrimination employment and occupation) Convention²⁰. All inhabitants of Kenya may turn to the UN Human Rights Committee, to the "Special Rapporteurs" for violations of specific human rights or to ECOSOC for women's rights violations.²¹ ICS can be purchased and used by any of the women within the project boundary willing to participate in the program. It will therefore not put at risk women's access to or control to efficient cookstoves. It's not foreseen either any reduction or risk related to any other resource, entitlement or benefit.
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²⁰ https://www.ilo.org/dyn/normlex/en/f?p=1000:11200:0::NO:11200:P11200_COUNTRY_ID:103315

²¹ <http://www.claiminghumanrights.org/kenya.html?L=0>

	Women/children (being the ones mostly spending time for cooking and fuel procurement) are able to spend less time on cooking/fuel procurement and cook in a much cleaner kitchen environment with less pollutants and dirt, resulting in health benefits and more time for income generating activities and education. The project will help to improve womens' health conditions as reduced combustion and less harmful gases during combustion will reduce indoor air pollution and thereby increase respiratory health of the women and children. Both women and men were participating in the local stakeholder consultation meeting. A strong focus has been put on women associations and groups when inviting stakeholders to the physical meeting. It is envisaged that women will be at the center of the project developer's marketing, education and distribution chain. It is planned that women in selected communities will demonstrate and educate consumers on the cookstove and its benefits resulting in empowerment, knowledge transfer and generation of jobs for women.
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Question 2 - Explain how the project aligns with existing country policies, strategies and best practices	The Project is in line with Kenya's constitution. Kenya has ratified the principle of equality into its respective constitution (Article 27), which shall guarantee equal gender rights. Item 3 of Article 27 mentions that "women and men have the right to equal treatment, including the right to equal opportunities in political, economic, cultural and social spheres". Article 21, item 3 mentions that "all State organs and all public officers have the duty to address the needs of vulnerable groups within society, including women, older members of society, persons with disabilities, children, youth, members of minority or marginalized communities, and members of particular ethnic, religious or cultural communities."²² It will be ensured that the project is committed to equal gender rights and women empowerment following Kenya's constitution.
Question 3 - Is an Expert required for the Gender Safeguarding Principles & Requirements?	Not applicable, since not necessary.

²² <http://kenyalaw.org/kl/index.php?id=398>

Question 4 - Is an Expert required to assist with Gender issues at the Stakeholder Consultation?	Not applicable, since not necessary.
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SECTION E. SUMMARY OF LOCAL STAKEHOLDER CONSULTATION

E.1. Summary of stakeholder mitigation measures

>>

The local stakeholder consultation (LSC) was conducted on 16, 17 and 19th May 2022. The detailed LSC report has been submitted to GS.

E.2. Final continuous input / grievance mechanism

Method	Include all details of Chosen Method (s) so that they may be understood and, where relevant, used by readers.	Justification
Continuous Input / Grievance Expression Process Book (mandatory)	BURN Manufacturing New Horizons Industrial Park Go Downs 8-11, Ruiru KENYA	The stakeholders were pleased to have an input book opened at the head office of BURN. Complaints or any other concerns can be stated and will be filed directly at the head office.
GS Contact (mandatory)	help@goldstandard.org	
Other	BURN: +254 706 585629 BURN: info@burnmfg.com	The phone number of the head offices of BURN will be made known to all project beneficiaries BURN's email address will be made known to all project beneficiaries

SECTION F. Eligibility and inclusion criteria for VPAs inclusion

>>

No.	Eligibility Criterion	Description/ Required condition	Description of the VPA in relation to the criteria, means of Verification/Supporting evidence for inclusion
1	Geographical Boundary	ICS distributed ²³ under any of the VPAs will be located in any of the countries mentioned under Table 1 of the PoA-DD.	ICS are distributed to urban and peri-urban households all over the territory of Kenya. For more details see section A.2. of this document.
2	Double-counting of project activities	All VPAs will be checked to prevent double counting and are not registered as a separate GS project activity, nor included as part of another registered GS (or other carbon standard) PoA nor that the project activity has been deregistered.	All carbon standard registries (UNFCCC, GS and VERRA) have been checked and it is confirmed that the VPA has not been registered as a separate GS project activity, nor included as part of another registered GS (or other carbon standard) PoA nor that the project activity has been deregistered. The same has been confirmed by a letter signed by the VPA implementer and submitted to GS.
3	Technology	Each VPA will implement improved biomass cook stoves.	The VPA implements highly efficient cookstoves known as 'Jikokoas'.

²³ Distributed' may include the free distribution of ICS, sale to full cost or subsidized cost.

		Detailed manufacturer's technology specifications are listed in section A.3 of this document.
4	Conditions to check the start date of the VPA through documentary evidence	The start date of a project activity is the date on which the first ICS has been distributed under the VPA. The start date of retroactive VPAs (with a start date prior to date of first submission of PoA) can be at the earliest 1 year prior to submission of documents for GS preliminary review. The start date is defined as 01/12/2020, i.e. the day when the first ICS is expected to be distributed to a household under this VPA. The start date of the VPA will be checked by sales agreement, electronic sales record, proof of shipping or any equivalent for the first ICS distributed under the VPA.
5	Methodology	Each VPA will comply with the applicability criteria of the applied methodology (TPDDTEC, version 03.1) The VPA complies with all applicability criteria of TPDDTEC as further outlined in section B.2 of this document.
6	Financial Additionality & Ongoing Financial Needs	All projects (VPAs) to be included under the PoA will be in compliance with item 1.1.3 of Annex B – positive list mentioned in the 'Community Services Activity Requirements'. All VPAs will be solely composed of isolated units (efficient cookstoves) where the users of the technology/measure are household/SMEs/institutions and where each unit results in $\leq 1,800$ MWh of thermal energy savings per year. The thermal energy savings per year at a unit level (i.e. per ICS) are clearly below 1,800 MWh as outlined in the ER calculation excel spreadsheet (worksheet 'Th. Energy savings unit level'). Hence, according to

paragraph 4.1.9 of the 'Community Services Activity Requirements', each of the VPAs, regardless of the host country in which the project activity is being implemented, is deemed additional and therefore is not required to prove financial additionality at the time of Design Certification.

7	Stakeholder inclusivity	Local stakeholder consultation is done at VPA level, as described in section F of the PoA-DD. Local stakeholder consultation report must be provided along with VPA-DD. A single Stakeholder consultation can be conducted for a group of VPAs as long as convincing justification is provided.	A local stakeholder consultation report for a group of VPAs has been submitted to GS. The physical LSC meeting and stakeholder feedback round have been conducted (see LSC report for more details). The Local Stakeholder Consultation would be also valid for any other Voluntary Project Activities (VPAs) implemented in Kenya under BURN's Gold Standard PoA 'ECO_A_BURN multi-country Clean Cooking Programme', provided that they are homogeneous, i.e. deploy the same stove type(s), target the same end-users and consist of the same project boundary as this specific VPA.
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8	Conditions related to environmental Impact Analysis	The VPA has to fulfil host country requirements (if host country for ICS any) concerning project activities as the environmental impact one described in this analysis. document ²⁴
9	CME approval	Each VPA has a project This eligibility criterion is implementer that is either not relevant for this VPA the since the CME is the same Coordinating/Managing entity as the VPA Entity or another entity implementer. that has signed a contractual agreement with the CME. Those agreements include all rights and responsibilities of both parties, e.g. approval procedures by the CME, monitoring requirements, carbon credit rights transfer. This eligibility criterion is not relevant if the CME is the VPA implementer.
10	Transfer of carbon credit ownership	The transfer of carbon credit ownership all along permanently waive any the investment chain is claim or rights on carbon clearly described and credits to the VPA communicated to all implementer (at the same project participants and time CME of the PoA). This end-users so that they are is confirmed by strap on aware of to give up their ICS box and warranty rights on emission booklet. reductions. For technology producers and the retailers There will be contractual of the improved agreements between technology, this must be distributors/retailers in communicated by contract which or clear written assertions distributors/retailers in the transaction waive any claim or rights

²⁴ See e.g. PoA9265, Appendix 8, https://cdm.unfccc.int/ProgrammeOfActivities/poa_db/BSVR8KUNAXW0LI1PGC6H42Y3JD9EQ7/vicw

		paperwork. The end-users on carbon credits to the will need to be informed VPA implementer (at the and notified that they same time CME of the cannot claim for emission PoA). The same will be reductions from the submitted to the VVB project. during the validation.
11	Conditions to provide an affirmation that funding from Annex I Parties, if any, does not result in a diversion of ODA	In case that any of the VPA The VPA implementer receives ODA, it is ensured has signed an ODA that there is no diversion declaration confirming of ODA, i.e. that no ODA is that there is no diversion provided under the of ODA. The same has condition that all or part of been submitted to GS. the carbon credits have to be returned to the donor country/entity providing ODA.
12	Target Group and distribution mechanism	The VPA serves This VPA targets households, institutions or households in urban and SMEs either in urban, peri-urban and/or rural areas across the entire territory of Kenya. and distributes the cook stoves through adequate distribution channels. ICS are distributed through direct sale/distribution and/or a variety of retail outlets across the country to end-users.
13	Conditions related to sampling requirements	The VPA complies with The VPA-DD outlines the the sampling plan as sampling plan in section outlined in the VPA-DD, B.7.2 which is in line with section B.7.2 the one stipulated in the PoA-DD. The VPA will make part of a single sampling covering a group of VPAs, as soon as there are other homogenous VPAs. In case of a grouped sampling approach, the CDM Project Standard for PoAs will be followed.
14	Double counting of emission reductions	Each VPA will implement The unique identification a unique identification system is explained in system for every efficient detail in section A.3. of cooking unit distributed this document. The VPA to avoid double counting is in adherence to the of emission reductions. CME Management

		System as outlined in Section C of the PoA-DD.
15	Crediting Period	The duration of the crediting period of the VPA does not exceed the end date of the registered PoA or shall be capped by the end date of the PoA. The final date for which ERs can be credited shall be no later than 20 years after the start date of the PoA. The VPA will have a crediting period of 5 years which can be renewed twice, i.e. in total a maximum issuance of 15 years. The VPA will not exceed the end date of the registered PoA.

APPENDIX 1 - SAFEGUARDING PRINCIPLES ASSESSMENT

Complete the Assessment below and copy all Mitigation Measures for each Principle into [SECTION D](#) above. Please refer to the instructions in the [Guide to Completing](#) this Form below.

Assessment Questions/ Requirements	Justification of Relevance (Yes/potentially/no)	How Project will achieve Requirements through design, management or risk mitigation.	Mitigation Measures added to the Monitoring Plan (if required)
Principle 1. Human Rights			
1. The Project Developer and the Project shall respect internationally proclaimed human rights and shall not be complicit in violence or human rights abuses of any kind as defined in the Universal Declaration of Human Rights 2. The Project shall not discriminate with regards to participation and inclusion	No	The project is implemented on the ground by BURN in collaboration with local distribution partners. The project developer takes care that the project respect internationally proclaimed human rights and is no complicit in violence or human rights abuses of any kind as defined in the Universal Declaration of Human Rights.	N/A

		Kenya has ratified many UN Human Rights Conventions.²⁵ 2. The project will not discriminate with regards to participation and inclusion as the improved cookstoves (ICS) can be purchased and used by everybody within the project boundary willing to participate in the program.	
Principle 2. Gender Equality			
1. The Project shall not directly or indirectly lead to/contribute to adverse impacts on gender equality and/or the situation of women 2. Projects shall apply the principles of non	No	The Project is in line with Kenya's constitution. Kenya has ratified the principle of equality into its respective constitution (Article 27), which shall guarantee equal gender	N/A

²⁵ <http://www.claiminghumanrights.org/kenya.html?&L=0>

discrimination, equal treatment, and equal pay for equal work 3. The Project shall refer to the country's national gender strategy or equivalent national commitment to aid in assessing gender risks 4. (where required) Summary of opinions and recommendations of an Expert Stakeholder(s)		rights. Item 3 of Article 27 mentions that "women and men have the right to equal treatment, including the right to equal opportunities in political, economic, cultural and social spheres". Article 21, item 3 mentions that "all State organs and all public officers have the duty to address the needs of vulnerable groups within society, including women, older members of society, persons with disabilities, children, youth, members of minority or marginalized communities, and members of particular ethnic, religious or cultural	
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		communities.” ²⁶ It will be ensured that the project is committed to equal gender rights and women empowerment following Kenya’s constitution.	
Principle 3. Community Health, Safety and Working Conditions			
1. The Project shall avoid community exposure to increased health risks and shall not adversely affect the health of the workers and the community	No	The project activity doesn’t expose the community to increased health risks and is not adversely affecting the health of workers and the community. Cooking with improved	N/A

²⁶ <http://kenyalaw.org/kl/index.php?id=398>

		cookstoves is actually safer than any other open flame stove use or traditional stoves. The workers participating in the project activity are not exposed to unsafe or unhealthy work environments as the sale/distribution of efficient cookstoves or the monitoring activities of the project will not include any hazardous chemicals or other hazardous material.	
Principle 4.1 Sites of Cultural and Historical Heritage			
Does the Project Area include sites, structures, or objects with historical, cultural, artistic, traditional or religious values or intangible forms of culture?	No	The project activity doesn't include sites, structures or objects with historical, cultural, artistic, traditional or religious values or intangible forms of culture. The Project	NA

>>		will introduce efficient cookstoves in urban and peri-urban households in the Republic of Kenya and it does not require alteration, damage or removal of any historical, artistic, traditional, religious or cultural heritage issues.	
Principle 4.2 Forced Eviction and Displacement			
Does the Project require or cause the physical or economic relocation of peoples (temporary or permanent, full or partial)?	No	The project activity will introduce portable improved cookstoves in urban and peri-urban households in Kenya and therefore no physical or economic relocation of people is involved. The use of efficient cookstoves is voluntarily.	NA
>>			
Principle 4.3 Land Tenure and Other Rights			
a.Does the Project require any change, or have any uncertainties related to land tenure arrangements and/or access rights, usage rights or land ownership?	No	The project doesn't require any change to land tenure arrangements and/or other rights.	NA

b. For Projects involving land use tenure, are there any uncertainties with regards to land tenure, access rights, usage rights or land ownership?		The project does not involve land-use tenure.	
>>			
Principle 4.4 - Indigenous people			
Are indigenous peoples present in or within the area of influence of the Project and/or is the Project located on land/territory claimed by indigenous peoples?	No	There are no indigenous people present within the area of influence nor the project has an impact on territory claimed by indigenous people.	NA
>>			
Principle 5. Corruption			
1. The Project shall not involve, be complicit in or inadvertently contribute to or reinforce corruption or corrupt Projects	No	The Project doesn't involve, be complicit in or inadvertently contribute to or reinforce corruption or corrupt Projects. The project is, in fact, implemented on the ground	NA

		by BURN. The ethical codes of BURN and other project partners are against corruption	
Principle 6.1 Labour Rights			
1. The Project Developer shall ensure that all employment is in compliance with national labour occupational health and safety laws and with the principles and standards embodied in the ILO fundamental conventions 2. Workers shall be able to establish and join labour organisations 3. Working agreements with all individual workers shall be documented and implemented and include:	No	The project is implemented on the ground by the enterprise BURN in collaboration with other project partners. The employees' rights are a cross-cutting issue and respected in all of the projects of BURN and other project partners. Kenya has ratified many ILO Conventions, amongst others convention 100 (Equal Remuneration Convention) and convention 98 (Right to Organise and Collective Bargaining	NA

a) Working hours (must not exceed 48 hours per week on a regular basis), AND b) Duties and tasks, AND c) Remuneration (must include provision for payment of overtime), AND d) Modalities on health insurance, AND e) Modalities on termination of the contract with provision for voluntary resignation by employee, AND f) Provision for annual leave of not less than 10 days per year, not		Convention).²⁷ All employees will work voluntarily for the project, no forced labour is used and all employment is in compliance with national laws and consistence with the principles and standards of the ILO conventions. In fact, Kenya has ratified many ILO Conventions, amongst others convention 29 (Forced Labour Convention) and 105 (Abolition of Forced Labour Convention).²⁸ The workers are able to establish and join labour	
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²⁷ https://www.ilo.org/dyn/normlex/en/f?p=1000:11200:0::NO:11200:P11200_COUNTRY_ID:103315

²⁸ https://www.ilo.org/dyn/normlex/en/f?p=1000:11200:0::NO:11200:P11200_COUNTRY_ID:103315

including sick and casual leave. 4. No child labour is allowed (Exceptions for children working on their families' property requires an Expert Stakeholder opinion) 5. The Project Developer shall ensure the use of appropriate equipment, training of workers, documentation and reporting of accidents and incidents, and emergency preparedness and response measures		organizations. The working agreements with the individual workers will be documented and implemented and the minimum requirements stated in section 3.6.1. of GS4GG Safeguarding Principles & Requirements (version 1.2) will be respected whenever applicable. The employment model applied will be also locally and culturally appropriate. Moreover, all the possible staff hired follows Convention C138 (Minimum Age Convention). Kenya has ratified besides C138 as well ILO Convention 182 (Worst	
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		Forms of Child Labour Convention). ²⁹ All the works will be made by using appropriate equipment, training of workers, documentation and reporting of accidents and incidents, and emergency preparedness and response measures.	
Principle 6.2 Negative Economic Consequences			
1. Does the project cause negative economic consequences during and after project implementation?	No	The project is based on a commercial model selling improved cookstoves in order to ensure the economic durability of the project. Carbon revenues are amongst others used for sensitizing end-users and awareness	NA
>>			

²⁹ https://www.ilo.org/dyn/normlex/en/f?p=1000:11200:0::NO:11200:P11200_COUNTRY_ID:103315

		raising, fortify the distribution/supply chain and upscale the project. Improved cookstoves can be purchased and used by everybody within the project boundary willing to participate in the program. There are not expected any direct economic impact or potential risks to the local economy.	
Principle 7.1 Emissions			
Will the Project increase greenhouse gas emissions over the Baseline Scenario?	No	The project will reduce the GHG emissions as it will be monitored and verified in line with the GS4GG.	NA
>>			
Principle 7.2 Energy Supply			
Will the Project use energy from a local grid or power supply (i.e., not connected to a national or regional grid) or fuel resource (such as wood,	No	The project does not use energy from a local grid or power supply. Biomass use (charcoal) will be significantly	NA

biomass) that provides for other local users?		reduced by introducing highly efficient charcoal stoves.	
>>			
Principle 8.1 Impact on Natural Water Patterns/Flows			
Will the Project affect the natural or pre-existing pattern of watercourses, ground-water and/or the watershed(s) such as high seasonal flow variability, flooding potential, lack of aquatic connectivity or water scarcity?	NO	The project activity will not negatively affect natural or pre-existing pattern of watercourses, ground-water and/or watersheds.	NA
>>			
Principle 8.2 Erosion and/or Water Body Instability			
a. Could the Project directly or indirectly cause additional erosion and/or water body instability or disrupt the natural pattern of erosion? b. Is the Project's area of influence susceptible to excessive erosion and/or water body instability?	No	The project will not cause additional erosion and/or water body instability of or disrupt the natural pattern of erosion.	NA

>>			
Principle 9.1 Landscape Modification and Soil			
Does the Project involve the use of land and soil for production of crops or other products?	No	The Project itself does not involve the use of land and soil for production of crops or other products.	NA
>>			
Principle 9.2 Vulnerability to Natural Disaster			
Will the Project be susceptible to or lead to increased vulnerability to wind, earthquakes, subsidence, landslides, erosion, flooding, drought or other extreme climatic conditions?	No	The Project will not be susceptible to or will lead to increased vulnerability to any extreme climatic conditions.	NA
>>			
Principle 9.3 Genetic Resources			
Could the Project be negatively impacted by or involve genetically modified organisms or GMOs (e.g., contamination, collection and/or harvesting, commercial development, or	No	The Project is not negatively impacted by the use of genetically modified organisms or GMOs.	NA

take place in facilities or farms that include GMOs in their processes and production)?			
>>			
Principle 9.4 Release of pollutants			
Could the Project potentially result in the release of pollutants to the environment?	No	The Project does not release any different pollutants to the environment which would not be released in the baseline already. The release of PM and carbon monoxide are significantly reduced by the introduction of efficient cookstoves.	NA
>>			
Principle 9.5 Hazardous and Non-hazardous Waste			
Will the Project involve the manufacture, trade, release, and/ or use of hazardous and non-hazardous chemicals and/or materials?	No	The Project is not involving the manufacture, trade, release, and/or use of hazardous chemicals and or materials.	NA
>>			
Principle 9.6 Pesticides & Fertilisers			

Will the Project involve the application of pesticides and/or fertilisers?	No		NA
>>			
Principle 9.7 Harvesting of Forests			
Will the Project involve the harvesting of forests?	No	No harvesting of forests is involved.	NA
>>			
Principle 9.8 Food			
Does the Project modify the quantity or nutritional quality of food available such as through crop regime alteration or export or economic incentives?	No	The Project doesn't modify the quantity or nutritional quality of food available.	NA
>>			
Principle 9.9 Animal husbandry			
Will the Project involve animal husbandry?	No	The Project doesn't involve animal husbandry.	NA
>>			
Principle 9.10 High Conservation Value Areas and Critical Habitats			

Does the Project physically affect or alter largely intact or High Conservation Value (HCV) ecosystems, critical habitats, landscapes, key biodiversity areas or sites identified?	No	The project is not located in an area within a high conservation value area or within critical natural habitats.	NA
>>		Furthermore, the aim of the project is to reduce biomass consumed in the project area for cooking which may save the natural resources.	
Principle 9.11 Endangered Species			
a. Are there any endangered species identified as potentially being present within the Project boundary (including those that may route through the area)? b. Does the Project potentially impact other areas where endangered species may be present through transboundary affects?	No	a. The project boundary is the physical, geographical sites of the distributed cookstoves. There are no endangered species identified as potentially being present the project boundary.	NA

>>		b. The distributed cookstoves are not expected to potentially impact other areas where endangered species may be present through transboundary affects.	
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APPENDIX 2- CONTACT INFORMATION OF VPA IMPLEMENTER

Organization name	BURN Manufacturing Co.
Registration number with relevant authority	Business ID: 603-137-240
Street/P.O. Box	Suite 220, 18850 103 rd Avenue
Building	
City	Vashon
State/Region	Washington
Postcode	98070
Country	United States
Telephone	+254 718 125 639
E-mail	peter.scott@burnmfg.com
Website	https://burnstoves.com
Contact person	Peter Scott
Title	
Salutation	Mr.
Last name	Scott
Middle name	
First name	Peter
Department	CEO
Mobile	
Direct tel.	
Personal e-mail	

APPENDIX 3- LUF ADDITIONAL INFORMATION

Risk of change to the Project Area during Project Certification Period:	
Risk of change to the Project activities during Project Certification Period:	
Land-use history and current status of Project Area:	
Socio-Economic history:	
Forest management applied (past and future)	
Forest characteristics (including main tree species planted)	
Main social impacts (risks and benefits)	
Main environmental impacts (risks and benefits)	
Financial structure	
Infrastructure (roads/houses etc):	
Water bodies:	
Sites with special significance for indigenous people and local communities - resulting from the Stakeholder Consultation:	
Where indigenous people and local communities are situated:	

Where indigenous people and local communities have legal rights, customary rights or sites with special cultural, ecological, economic, religious or spiritual significance:

APPENDIX 4-SUMMARY OF APPROVED DESIGN CHANGES

Please refer to [Design Changes Requirements](#) for more information on procedures governing Design Changes

Revision History

Version	Date	Remarks
2.0	4 May 2022	
1.1	7 October 2020	Hyperlinked section summary to enable quick access to key sections Improved clarity on Key Project Information Inclusion criteria table added Gender sensitive requirements added Prior consideration (1 yr rule) and Ongoing Financial Need added Safeguard Principles Assessment as annex and a new section to include applicable safeguards for clarity Improved Clarity on SDG contribution/SDG Impact term used throughout Clarity on Stakeholder Consultation information required Provision of an accompanying Guide to help the user understand detailed rules and requirements
1.0	10 July 2017	Initial adoption